

SHIVA SHANKAR S

AI Engineer · RAG Systems · LLM Agent Architecture · LangChain · LangGraph

s.shivashankar1304@gmail.com | +91 7676823681 | github.com/Shiva7147 | Bengaluru, India — Open to Remote / Relocating to Spain

PROFESSIONAL SUMMARY

AI engineer who builds and ships working AI systems, not just experiments. Hands-on experience designing production RAG pipelines, LangGraph ReAct agents with tool-calling, and automated backend systems deployed to real infrastructure. Understands the full stack from LLM internals (tokens, embeddings, attention) through deployment. Every project evaluated with documented accuracy metrics rather than demoed and forgotten. Open to AI/ML engineering internships and remote roles, with a longer-term focus on the Spanish and European market.

TECHNICAL SKILLS

AI/LLM Engineering: RAG pipelines, LangChain LCEL, LangGraph ReAct agents, ChromaDB, FAISS, prompt engineering, system prompt design, tool calling, LLM evaluation (LLM-as-judge + rule-based), hybrid eval harnesses

LLM APIs & Models: OpenRouter, Anthropic Claude, OpenAI API, Google Gemini, HuggingFace embeddings (all-MiniLM-L6-v2), multi-model routing

Agent Architecture: ReAct pattern, tool design with @tool decorator, multi-tool orchestration, safety guardrails, tiered escalation systems, agentic loop management

Languages & Core: Python, TypeScript, JavaScript, SQL, Git/GitHub, REST APIs

Backend & Infra: Node.js, Fastify, Next.js, Supabase, PostgreSQL, Railway, Firebase, Twilio Voice

Data & Automation: Pandas, data cleaning/validation, structured reporting, workflow automation

Responsible AI: EU AI Act awareness, safety-first agent design, human-in-the-loop escalation, bias mitigation, AI system evaluation

Also Learning: LangGraph multi-agent systems, Langfuse observability, Docker, Google Cloud Run

PROJECTS

AthleteIQ — AI Sports Science RAG Agent github.com/Shiva7147/athleteiq

- Built a production-grade LangGraph ReAct agent with four specialized tools: sports-science RAG retrieval over ChromaDB, an automated ACWR injury-risk calculator (deterministic Python math, not LLM prediction), timestamped athlete metric logging, and a three-tier safety escalation system.
- Designed a 56-entry evidence-based knowledge base covering ACWR, HRV, RPE, overtraining, biomechanics, and youth athlete safety, embedded with HuggingFace all-MiniLM-L6-v2 and stored in ChromaDB for semantic retrieval.
- Built an end-to-end evaluation harness across 20 ground-truth test cases in 9 categories, combining LLM-as-judge scoring with rule-based checks for safety-critical questions — improving measured accuracy from 85% to 95% through targeted knowledge base additions and refined scoring.
- Implemented Responsible AI guardrails: the agent skips knowledge retrieval entirely for medical emergencies and escalates immediately, aligned with EU AI Act principles for human oversight in health-adjacent AI systems.

Tech: Python, LangChain LCEL, LangGraph, ChromaDB, HuggingFace, OpenRouter API

Alpha Studio AI Receptionist *Production AI Voice Automation Platform*

- Built a production-ready AI voice receptionist that handles inbound business phone calls in real time using Twilio Voice, integrated with Google Gemini for natural, sub-second-latency conversations.
- Developed a business-specific knowledge base for company-aware conversations, plus automated call transcription, AI summarization, and lead qualification.
- Built a cloud-hosted backend (Node.js, Fastify, Railway, Supabase/PostgreSQL) with a dashboard for call analytics, transcripts, and customer insights, and integrated Google Calendar for AI-powered appointment scheduling.
- Architected the platform to support future multi-tenant SaaS deployment.

Tech: Next.js, React, TypeScript, Node.js, Fastify, Supabase, PostgreSQL, Twilio, Google Gemini, Railway

Reframe — AI Habit Recovery Web App *github.com/Shiva7147/reframe-app*

- Built a full-stack Next.js and Gemini 2.0 Flash web app for habit-based behavioral support, including a multi-dimensional weighted assessment algorithm, personalized onboarding, a real-time urge-interrupt feature, and progress visualization with Recharts.
- Implemented client-side AES-GCM encryption with PBKDF2 key derivation, anonymous Firebase authentication, and offline-first PWA support with automatic sync.

Tech: Next.js, Gemini 2.0 Flash, Firebase Firestore, Tailwind CSS, Recharts, Vercel

Smooth Operator — AI Dating Profile Coach *Founder*

- Founded and designed Smooth Operator, a platform concept connecting men on dating apps with real reviewers who provide personalized voice-note feedback on their profiles, structured across three pricing tiers.
- Designed the AI layer architecture: a RAG knowledge base of dating archetypes and conversation templates, agent-based profile analysis, and a reviewer feedback retrieval system for personalized coaching.
- Produced complete go-to-market assets, including ad campaign briefs, outreach content, and a partnership outreach list with media kit.

Machine Unlearning in LLMs *IEEE Generative AI Challenge — Research Paper*

- Co-authored “Surgical Amnesia: A Framework for Targeted Machine Unlearning and Copyright Compliance in Large Language Models,” presenting a parameter-efficient unlearning method (Llama-3.2-1B, LoRA with a negated cross-entropy objective) to meet GDPR and EU AI Act data-erasure requirements without full retraining.
- Team reached the final round of the IEEE Industrial Electronics Society Generative AI Challenge. Revised methodology per reviewer feedback and submitted in IEEE format.

EDUCATION

Master of Computer Applications (MCA) — St Joseph's University, Bengaluru 2025 – 2027 (Expected)

Focus: AI/ML engineering, RAG systems, agentic architectures, LLM evaluation.

Bachelor of Computer Applications (BCA) — Kristu Jayanti University 2021 – 2024

Internships in web development and data analysis. Coursework in big data analytics: Hadoop, MapReduce, NoSQL, Hive, Pig.